

Scalar-Carrier Shell-Native Current-Closure Release for HAOS-IIP

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Abstract

This paper freezes release 52.4 of HAOS-IIP as the bounded shell-native follow-on to the open transient current probe 52.3. The carrier is unchanged: the same local scalar kernel graph, the same bounded local metric field from 51.4, and the same clean refinement line. What changes is only the transient reconstruction. Release 52.3 compared cumulative shell-mass flow against a constitutive law that still mixed shell-center and shell-boundary viewpoints, and the fitted conductivity scaled incorrectly with refinement. Release 52.4 keeps the empirical cumulative shell current but rebuilds the constitutive side on exact bulk shells and shell density $\rho = m_{\text{shell}}/(N_{\text{shell}}h^3)$. On this corrected shell-native read, all three clean refinement cases pass. The fitted conductivity stays close to the heat diffusivity, the relative residuals remain small, and the normalized shellwise constitutive profile stays stable under refinement. The correct bounded conclusion is therefore positive but narrow: on the clean scalar carrier, a shell-native transient current reconstruction closes onto one bounded constitutive family. This is a transport-law closure on the same scalar geometry carrier, not yet a disorder-universal current law, conserved-current theorem, curvature result, or spacetime claim.

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1 Scope

Release 52.4 is a narrow scalar follow-on release.

It does not add:

- disorder or kernel-family universality for the transient law;
- a conserved-current theorem;
- curvature extraction;
- spacetime or ontology language;
- any new carrier beyond the already validated scalar line.

It does add:

- a corrected shell-native transient current reconstruction on the same carrier;
- a bounded constitutive family whose fitted conductivity tracks the heat diffusivity on the clean refinement line;
- a clean separation between the open 52.3 baseline and the corrected 52.4 closure.

2 Where 52.3 Left the Boundary

Release 52.3 asked the right stronger question after the positive static 52.2 inverse-square closure:

does the same scalar carrier also support a constitutive response/current closure?

Its answer was boundedly open. The main symptoms were:

- median relative error around 0.49 to 0.59;
- p_{90} relative error around 0.60 to 0.67;
- normalized current-profile drift above the bounded threshold;
- fitted conductivity scaling incorrectly with refinement.

The important point is what *did not* fail:

- the scalar carrier;
- the 51.4 local metric field;
- the positive 52.2 shell-native inverse-square response law.

So the next honest step was not to abandon current closure, but to correct the transient reconstruction itself.

3 What 52.4 Changes

Release 52.4 keeps the carrier fixed:

- same local scalar kernel graph;
- same coarse local metric field;
- same clean Euclidean shell scaffold;

- same clean refinement line $n = 13, 15, 17$.

The only change is the reconstruction:

- use exact bulk shells on the clean scaffold rather than a looser shell-profile surrogate;
- keep empirical current as cumulative shell-mass flow;
- read the constitutive law against shell density

$$\rho(r, t) = \frac{m_{\text{shell}}(r, t)}{N_{\text{shell}}(r) h^3};$$

- compare both sides through

$$I_{\text{const}}(r, t) = -\kappa 4\pi r^2 g_{rr}(r) \partial_r \rho(r, t).$$

The closure diagnostic is therefore not the raw current profile alone. It is whether one bounded conductivity family:

- stays close to the heat diffusivity;
- keeps small relative residuals;
- remains stable across shell radius and refinement.

4 Frozen Quantitative Summary

Case	D_{eff}	κ_{fit}	κ/D_{eff}	median err.	$p90$ err.	shell CV
n=13	1.0593	0.9397	0.8871	0.0705	0.2328	0.0967
n=15	1.0466	0.9530	0.9106	0.0709	0.1984	0.0742
n=17	1.0368	0.9604	0.9263	0.0739	0.2718	0.1176

Table 1: Bounded shell-native transient current-closure summary frozen by release 52.4.

The refinement-stability read is also positive:

- normalized shell- κ drift $13 \rightarrow 15 = 0.0493$;
- normalized shell- κ drift $15 \rightarrow 17 = 0.0966$.

5 What 52.4 Now Supports

The strongest honest statement now supported by the repository is:

on the clean scalar carrier, a shell-native transient current reconstruction closes onto one bounded constitutive family whose fitted conductivity stays close to the heat diffusivity and whose shellwise profile remains refinement-stable.

This matters because the scalar line now supports not only:

- one shared-family geometry bridge;
- one bounded local metric field;
- one shell-native inverse-square response law;

but also:

- one bounded transient transport-law closure on the same carrier.

That is the closest current scalar result to an effective physics-style constitutive law in the repository.

6 What 52.4 Still Does Not Say

Release 52.4 still does not justify:

- a disorder-robust or kernel-family-robust transient current law;
- a conserved-current theorem;
- curvature extraction;
- spacetime or ontology language;
- a universal transport law.

So 52.3 remains the honest open baseline for the first transient probe, while 52.4 is the corrected shell-native clean-line closure.

7 The Right Next Move

The next honest scalar step should remain narrow:

- test controlled inhomogeneity on the same carrier;
- then test mild disorder and bounded kernel-family variation for the transient law itself;
- only afterward ask for conserved-current or curvature-style closure.

8 Conclusion

Release 52.4 freezes the corrected shell-native transient current closure on the clean scalar carrier.

This is not a claim that all current-like questions are solved. It is a narrower and more useful statement: once the transient law is reconstructed on exact bulk shells and compared through shell density rather than raw node mass, the same scalar carrier supports one bounded constitutive family whose fitted conductivity remains close to the heat diffusivity and whose shellwise profile remains refinement-stable. That is the correct stopping point for 52.4: stronger than the open 52.3 baseline, still disciplined about what has not yet been earned.